

# PYNQ-Z2 Features

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## PYNQ

PYNQ is an open-source project from Xilinx that makes designing embedded systems with Xilinx Zynq SoCs easy. Using Python and libraries, designers can exploit the benefits of programmable logic and microprocessors in Zynq to build more capable and exciting embedded systems. PYNQ users can create high-performance embedded applications with parallel hardware execution, high frame-rate video processing, hardware-accelerated algorithms, real-time signal processing, high bandwidth IO, and low latency control.

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## Features

- [ZYNQ XC7Z020-1CLG400C](#)
  - 650MHz Arm® Cortex®-A9 Dual-core Processor
  - Programmable logic
    - 13,300 logic slices, each with four 6-input LUTs and 8 flip-flops
    - 630KB block RAM
    - 220 DSP slices
    - On-chip Xilinx analog-to-digital converter (XADC)
  - Programmable from JTAG, Quad-SPI flash, and MicroSD card
- Memory and storage
  - 512MB DDR3 with 16-bit bus @ 1050Mbps
  - 16MB Quad-SPI Flash with factory programmed 48-bit globally unique EUI-48/64™ compatible identifier
  - MicroSD slot
- Power
  - USB or 7V to 15V external power regulator
- USB and Ethernet
  - Gigabit Ethernet PHY
  - Micro USB-JTAG programming circuitry
  - Micro USB-UART bridge
  - USB 2.0 OTG PHY (supports host only)
- Audio and Video
  - 2x HDMI ports (input and output)
  - 24-bit I2S DAC with 3.5mm TRRS jack
  - Line-in with 3.5mm jack
- Switches, Push-Buttons, and LEDs
  - 4x push-buttons
  - 2x slide switches
  - 4x LEDs
  - 2x RGB LEDs
- Expansion Connectors
  - 2x Pmod ports

- 16 Total FPGA I/O (8 pins on Pmod A are shared with Raspberry Pi connector)
- Arduino Shield-compatible connector
  - 24 Total FPGA I/O
  - 6 Single-ended 0V to 3.3V Analog inputs to XADC
- Raspberry Pi connector
  - 28 Total FPGA I/O (8 pins are shared with Pmod A)
- Board dimensions: 87mm x 140mm (3.43" x 5.51")